

# Multi-Agent AI Engineering Curriculum

A self-study curriculum for building production-grade multi-agent AI systems — 19 documents and 6 real projects, from Python fundamentals to a deployed, tested, multi-agent platform.

This is not a theory course. Every document pairs its concepts with hands-on practice exercises (each with progressive hints and full worked solutions), and every project is real, runnable code you build yourself, step by step.

## How this curriculum is structured

- **19 documents** ( `01_python_foundations/` → `19_mlops_llmops/` ) — each one teaches a topic through simple-English "Core Concepts," then a set of practice exercises with a 2-hint, 3-depth (Basic/Intermediate/Advanced) hint system and full solutions, then a Build Task that feeds directly into the projects below.
- **6 projects** ( `project_1_beginner_llm_app/` → `project_6_codeguard_bonus/` ) — real multi-agent systems you build across the documents, growing from a 2-agent terminal app to a tested, deployed, production-ready platform.
- Every document and project is delivered as both a `README.md` (source) and a matching `README.pdf` (for comfortable reading), with fully clickable internal links between exercises, hints, and solutions.

## Documents

| #  | Document                                              | Feeds into                 |
|----|-------------------------------------------------------|----------------------------|
| 01 | <a href="#">Python &amp; Engineering Foundations</a>  | —                          |
| 02 | <a href="#">APIs, HTTP, JSON &amp; Backend Basics</a> | —                          |
| 03 | <a href="#">LLM Basics</a>                            | —                          |
| 04 | <a href="#">OpenAI API</a>                            | Project 1                  |
| 05 | <a href="#">LangChain Basics</a>                      | —                          |
| 06 | <a href="#">Tools &amp; Function Calling</a>          | Projects 7, 8 (MCP topics) |
| 07 | <a href="#">AI Agents</a>                             | Project 2                  |
| 08 | <a href="#">RAG</a>                                   | —                          |
| —  | <a href="#">Async Python (checkpoint)</a>             | —                          |

| #  | Document                                                | Feeds into |
|----|---------------------------------------------------------|------------|
| 09 | <a href="#">LangGraph</a>                               | —          |
| 10 | <a href="#">Agent Workflows</a>                         | Project 3  |
| 11 | <a href="#">Multi-Agent Systems</a>                     | Project 4  |
| 12 | <a href="#">Production AI Engineering</a>               | —          |
| 13 | <a href="#">Testing, Evaluation &amp; Observability</a> | —          |
| 14 | <a href="#">Debugging Lab</a>                           | —          |
| 15 | <a href="#">Five Projects (reference index)</a>         | —          |
| 16 | <a href="#">System Design &amp; Architecture</a>        | —          |
| 17 | <a href="#">Interview Preparation</a>                   | —          |
| 18 | <a href="#">Final Production Multi-Agent Capstone</a>   | Project 5  |
| 19 | <a href="#">MLOps &amp; LLMOps</a>                      | —          |

## Projects

| #            | Project                                                      | Agents                  | Built during                     | Proves                                                                                 |
|--------------|--------------------------------------------------------------|-------------------------|----------------------------------|----------------------------------------------------------------------------------------|
| 1            | <a href="#">SupportDesk: Concierge &amp; Triage</a>          | 2                       | Docs 01-04                       | You can talk to a model correctly and safely, and route between two simple agents      |
| 2            | <a href="#">ResearchHand: Worker &amp; Verifier</a>          | 2                       | Docs 05-07                       | You can give a model real abilities, check its choices, and add an independent check   |
| 3            | <a href="#">DocuMind: Retriever, Reasoner &amp; Approval</a> | 3                       | Docs 08-10                       | You can build agents as a clear, checkable graph, with real handoffs                   |
| 4            | <a href="#">ContentForge: Supervisor-Led Team</a>            | 5                       | Doc 11                           | You can split a problem across specialist agents, with a supervisor routing on the fly |
| 5            | <a href="#">ContentForge Pro: Production Platform</a>        | 5<br>(production-ready) | Docs 12-19                       | You can ship, test, secure, watch, version, and defend the design of #4                |
| 6<br>(bonus) | <a href="#">CodeGuard: PR-Review Team</a>                    | 4                       | All of Docs 01-19, applied again | You can apply the whole architecture to a new problem, with less help                  |

| #            | Project                                     | Agents                 | Built during                  | Proves                                                                                                |
|--------------|---------------------------------------------|------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------|
| 7<br>(bonus) | <a href="#">MCPForge: MCP Server</a>        | — (server, not agents) | Doc06's MCP topics            | You can expose real tools/resources/prompts over the Model Context Protocol, usable by any MCP client |
| 8<br>(bonus) | <a href="#">MCPBridge: MCP Client Agent</a> | 1 (dynamic tools)      | Doc06/07's MCP + ReAct topics | You can build an agent whose tools are discovered at runtime from live MCP servers, not hardcoded     |

Agent count grows deliberately across the arc (2 → 2 → 3 → 5 → 5) — each project adds one new idea on top of the last, not several at once.

## How to use this

1. Start at [01\\_python\\_foundations/](#) and work through the documents in order — each one lists its prerequisites at the top.
2. Read a document's Core Concepts, then do its Practice Exercises. Try each one yourself before opening its hints — use Hint 1, then Hint 2 only if you're still stuck, and the full solution only as a last resort.
3. When a document's Build Task is done, check [15\\_five\\_projects\\_index/](#) and start the project it unlocks.
4. Track your own progress in [PROGRESS.md](#) .
5. See [CURRICULUM.md](#) for the full syllabus, rationale, and document-by-document breakdown.

## Building the PDFs

Every `README.md` has a matching `README.pdf` with clickable internal links. If you edit any `.md` file, regenerate its PDF with the build tool in `tools/` :

```
cd tools
python3 build_pdfs.py                # rebuild everything
python3 build_pdfs.py ../01_python_foundations/README.md  # rebuild just one file
python3 check_links.py               # verify every internal link resolves correctly
```

`watch_pdfs.py` can also be left running to rebuild automatically whenever a `.md` file changes.